

Some key results (to learn):

$$\int \sin x dx = -\cos x + c \quad \int \cos x dx = \sin x + c$$

$$\int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + c$$

$$\int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + c$$

$$\int \sec^2 x dx = \tan x + c$$

C4 Integration

You need to remember these trigonometric formulae- they are needed in some integration questions.

$$\sin^2 x \equiv \frac{1}{2}(1 - \cos 2x) \quad \cos^2 x \equiv \frac{1}{2}(1 + \cos 2x)$$

$$\sin^2 x + \cos^2 x = 1 \quad \tan^2 x + 1 = \sec^2 x$$

$$1 + \cot^2 x = \operatorname{cosec}^2 x$$

Integration by substitution

This column gives the calculations for changing the dx to du :

Example: $\int x(x^2 + 3)^5 dx$.

Make the substitution $u = x^2 + 3$

We get: $\int x u^5 dx = \int \frac{1}{2} u^5 du$

This gives: $\frac{1}{12} u^6 + c = \frac{1}{12} (x^2 + 3)^6$

$$u = x^2 + 3$$

$$\frac{du}{dx} = 2x$$

$$du = 2x dx$$

$$\text{So, } x dx = \frac{1}{2} du$$

Example 2: Use the substitution

$t = 2x + 1$ to find $\int \frac{x+1}{\sqrt{2x+1}} dx$.

Solution: $\int \frac{x+1}{\sqrt{t}} dx = \int \frac{1}{2} \frac{x+1}{\sqrt{t}} dt$

Since $t = 2x + 1$,

$$2x = t - 1 \Rightarrow x = \frac{1}{2}(t - 1)$$

So we get $\int \frac{1}{2} \frac{\frac{1}{2}(t-1)+1}{\sqrt{t}} dt$

Take the multipliers outside the integral:

$$\frac{1}{4} \int \frac{(t-1)}{\sqrt{t}} dt = \frac{1}{4} \int (t-1)t^{-1/2} dt = \frac{1}{4} \int t^{1/2} - t^{-1/2} dt$$

This gives: $\frac{1}{4} \left(\frac{2}{3} t^{3/2} - 2t^{1/2} \right) + c$

But $t = 2x + 1$, so $\frac{1}{4} \left(\frac{2}{3} (2x+1)^{3/2} - 2(2x+1)^{1/2} \right) + c$.

This expands to give: $\frac{1}{6} (2x+1)^{3/2} - \frac{1}{2} (2x+1)^{1/2} + c$

Methods of integration

The methods of integrating an expression are:

- 1) directly writing it down (either by memory or by looking in the formula book);
- 2) writing the expression in partial fractions;
- 3) using the method of integration by parts;
- 4) using a substitution;
- 5) using a trigonometric identity (such as $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$ or $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$)

See my separate sheet on how to identify which method of integration is the appropriate one to use in any given situation.

Definite integrals using a substitution

Find $\int_0^2 4xe^{x^2} dx$.

Use the substitution $u = x^2$.

$$\begin{aligned} \int_0^2 4xe^{x^2} dx &= \int_{u=0}^4 2e^u du = \left[2e^u \right]_0^4 \\ &= 2e^4 - 2e^0 = 2e^4 - 2 \end{aligned}$$

$$u = x^2$$

$$\frac{du}{dx} = 2x$$

$$du = 2x dx$$

$$4x dx = 2 du$$

$$x = 0 \rightarrow u = 0$$

$$x = 2 \rightarrow u = 4$$

Integration involving trigonometrical identities

Example: Using the identity given above

$$\int \cos^2 x dx = \int \frac{1}{2}(1 + \cos 2x) dx = \frac{1}{2} \left(x + \frac{1}{2} \sin 2x \right) + c$$

Example 2: Find $\int \sin^3 x dx$

Solution: We use the fact that $\sin^2 x = 1 - \cos^2 x$

$$\begin{aligned} \text{So, } \int \sin^3 x dx &= \int \sin^2 x \sin x dx = \int (1 - \cos^2 x) \sin x dx \\ &= \int (\sin x - \sin x \cos^2 x) dx \\ &= \int \sin x dx - \int \sin x \cos^2 x dx \end{aligned}$$

We now need to work out $\int \sin x \cos^2 x dx$

We use the substitution

$$u = \cos x \Rightarrow \frac{du}{dx} = -\sin x \Rightarrow du = -\sin x dx$$

$$\begin{aligned} \text{So, } \int \sin x \cos^2 x dx &= \int -u^2 du = \frac{-u^3}{3} \\ &= -\frac{1}{3} \cos^3 x \end{aligned}$$

$$\begin{aligned} \text{Therefore, } \int \sin^3 x dx &= -\cos x - \left(-\frac{1}{3} \cos^3 x \right) + c \\ &= -\cos x + \frac{1}{3} \cos^3 x + c \end{aligned}$$